

PADDY LEAF DISEASE DETECTION USING IMAGE PROCESSING*

BY

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Farmers need automatic disease monitoring of plants rather than human monitoring to increase crop field growth and yield. Manual disease monitoring is poor because it requires an expert to recognise disease, which takes longer and demands more time, making it ineffective. As a result, this research introduced a new technique for detecting illness in both leaves and fruits. A computer image processing technique was utilised to identify plant disease rapidly and effectively, overcoming the limits of traditional visual observation. Using a k-means clustering method and a multi SVM technique, we developed software for disease diagnosis and classification in our suggested study. This suggested method gives an overview of how machine learning might be used to classify and identify plant leaf diseases. Three- or more-layer neural networks are one subclass of machine learning approaches based on an artificial neuron model.

KEYWORDS

Plant Diseases detection using MATLAB, Multi SVM Algorithm, Image Pre-processing.

1. INTRODUCTION

As a result of infrequent plant inspections, the plant develops many diseases that must be treated with additional chemicals. Insects, birds, and other helpful species can also be poisoned by these toxins. The early detection of disease symptoms on a plant's growing leaf or fruit requires automatic detection of plant ailments. This paper describes a MATLAB-based system that employs image processing techniques to reliably identify and diagnose plant illnesses. An information system based on sensors in the yield field will be developed and implemented to offer executives with PDA and web application-based reports on the system's status. Equipment, web, and a versatile application are the three components. Furthermore, this

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paradigm emphasises the importance of computerised development in agribusiness. To enhance the image's contrast, various techniques such as histogram equalisation and contrast management are applied. K-means Processing approaches include FCM, PCA, PSO, and MPSO. The image displays one or more features in the classification algorithm's assumption. SVM, ANN, and fuzzy classification are all examples of classification characteristics. Texture features, structural features, and geometric characteristics are all examples of feature extraction. *Alternaria alternata*, a fungus, causes significant damage to the leaf. Greyish, brown patches emerge around the tip and border of immature leaves as the disease progresses. Leaf curl, mortality, and defoliation come from the lesions that extend towards the midrib.

Using python or java, a disease is diagnosed with its name and cures and shown on the app based on the selected image. After a disease is identified, farmers take the proper action, such as turning on or off the sprinkler assembly via an app, spraying pesticides or fertiliser by mixing it with water, and so on. They are not alone in this. A single pole double throw relay and a relay driver are used to turn on and off external devices. The sensor allows the farmer to monitor both the quality of the soil and the volume of water in the tank. Four different types of sensors are used to gauge soil health and water or pesticide tank levels.

2. LITERATURE REVIEW

Chili plant disease detection using MATLAB. A technique that combines two softwares makes it possible to perform an early leaf inspection. MATLAB was used to perform the picture processing. Edge detection, fourier filtering, and morphological procedures are used in image pre-processing, and colour clustering is used to identify chilli and non-chilli leaves. Chili plant health is demonstrated through image recognition and classification. Blast is a paddy disease caused by the fungus *Fungus magnaporthe oryzae*. Leaves and other parts of the panicle's neck are also affected by a blast. To use the image taken to identify disease in paddy leaves. The image distortion is eliminated after pre-processing and segmentation via spastically region merging. The RGB image is transformed to a binary image using an automated threshold based on the entropy method. The crop size and area are determined using the maximum likelihood approach. Neural network approaches were used to identify disease classification and detection. Jinup Yuan identified the SAR-processed region by using land image processing. The general view of this project is depicted in Figure 1. The PIC microcontroller receives the results of the photo processing. This makes a judgement based on the user's input and sends an SMS to the user's mobile phone. As demonstrated in Fig. 1, The various sensor blocks aid in the measurement of soil characteristics. The user could see information about the temperature,

humidity, and moisture content of the soil. When it is time to transmit the data to the user, GSM is employed.

3. PROPOSED METHODOLOGY

The proposed system's block diagram is presented in fig. 2 below. SVM methodology is then used to train the system on the database of leaf and fruit images that have been pre-processed, segmented using k-means clustering, and feature extracted using GLCM.

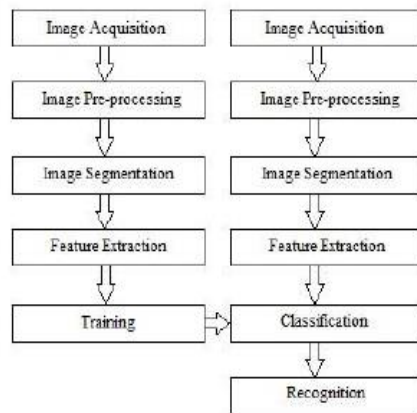


Fig. 2 Planned approach to implementation

3.1 Image Acquisition

Images of paddy leaves are taken using the image capture process. Using a digital camera, we were able to capture paddy leaf photos in RGB colour. Sample photos with a pixel resolution of 109*310 and a file size of roughly 225kb can also be found on the internet. Images are saved in bmp or jpeg format using the MATLAB image processing library. This is the first step of digital imaging processing, which is described as photographing an object with a digital camera and saving it to a computer for later use. It's also the process of extracting an image from a computer's hard drive for further processing. Digital photographs of leaf and fruit were taken using a digital camera for use in MATLAB image processing, as seen in fig 3.



Fig. 3 Infected leaf and berries in original image

Image Pre-processing

The method's main goal is to improve the image data by resizing it. Image scaling is required for image display, storage, and transmission. Picture enhancement is the outcome of image pre-processing, and the paddy leaf image now has dimensions of 109*310 pixels. It's in the RGB colour scheme. Many approaches are used in the preprocessing stage include scaling and shaping, noise filtering, image conversion and improvement as well as various morphological procedures and methods of pre-processing As illustrated in fig. 4, various MATLAB code was used to enlarge the image, boost the contrast, and convert RGB to grayscale. For subsequent actions, such as segmentation cluster creation.



Fig.4 shows the image pre-processing
Image segmentation

Using image segmentation, digital images can be broken down into smaller parts and analysed more readily. Image segmentation is used to locate an image's object and border lines. Clusters were created using the K-means clustering method, with at least one image in each one showing the sick part's main region. The image quality is harmed by camera flash, which creates noise, which is reduced with the k-means algorithm. From a binary image with noise, a noise-free image is created. Filtered photos are images that are devoid of noise. The k-means algorithm is used to create an enhanced image from a filtered image.

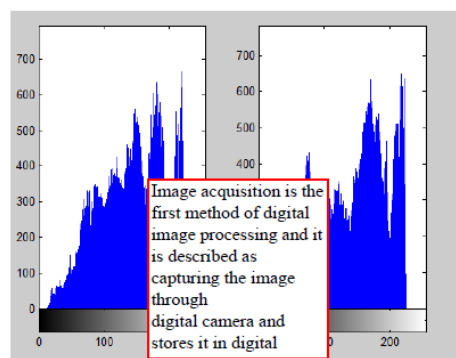


Fig. 5, Image Acquisition

Feature Extraction

Extracting feature vectors such as colour, texture, morphology, and structure is the goal of feature extraction. Reduce the resources needed to accurately characterise a large quantity of data by using feature extraction. The Gray level co-occurrence matrix (GLCM) approach for

texture analysis generates texture characteristics based on the statistical distributions of observed intensity combinations at the specified site compared to other points in the scene. The number of grey levels is significant in GLCM, and statistics for the number of intensity points in each combination are classed in order of first, second, and higher. GLCM's statistical texture qualities include energy, sum entropy, covariance, information measure of correlation, entropy, contrast, and inverse difference and difference entropy.

Support Vector Machine

SVM is a supervised machine learning classification model. It is possible to use SVM to solve linear and non-linear problems with the same technique. Afterwards, categorization is done by finding the hyper-plane that most clearly separates the two classes. SVM, a binary classifier, uses the decision boundary hyperplane between two classes. Different classes can be assigned to data points on either side of the hyper-plane. The hyperplane's size is determined by the number of features. It will use the training vectors to find the hyperplane that maximises the separation between the two classes, as depicted in Fig. 6.

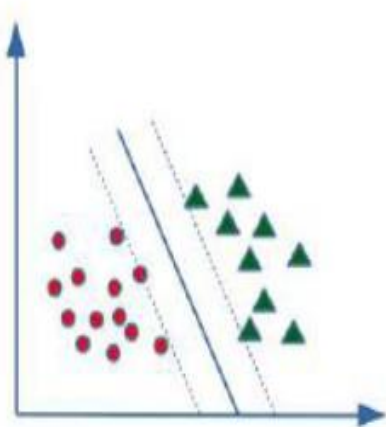


Fig 6: SVM in linearly separable conditions

Future Work

Image acquisition and conversion of RGB images to grayscale images are the techniques used. The grayscale images are then converted to binary images with noise. In addition, to eliminate the noise, the morphological algorithm employs region filling techniques. The paddy disease is identified with a 94.7 percent accuracy rate. The crop leaf disease can be detected automatically using this technology. Using digital image processing, an automatic technique for identifying and measuring paddy leaf disease signs was presented. A approach for identifying the symptoms of the paddy leaf is to use a traditional colour digital

photograph. This approach was presented for determining disease severity by automatically removing the risk of human errors and subsequently minimising the time. developed an approach for identifying plant diseases in large-scale crop monitoring Identifying paddy leaf disease by using a mobile application to capture photographs on smart phones. Preprocessing and segmentation using the stastical region merging (SRM) approach are used to eliminate noise from the images.

CONCLUSION

This work established a method for diagnosing paddy problems that is primarily based on Matlab applications. The classification method was created using image processing techniques. The classification system was developed using four characteristics: broken paddy leaf and lesion kind are also included in this list. Using the height/width ratio of the lesion location, researchers were able to identify the type of lesion present. Based on K-means and Multi SVM algorithms, the proposed methodology in this paper is able to detect both leaf and fruit illnesses. The MATLAB software is perfect for digital image processing. K-means clustering with the SVM algorithm deliver high accuracy while consuming relatively little processing time. We hope to expand our database in the future to include more plant disease identification. Consider the dataset and the quantity of photographs in the database when designing the application.

REFERENCES

- Channamallikarjuna Mattishall, Edemiale Gedefaye, Fasil Endalamaw, "Plant leaf diseases detection and auto-medicine" Internet of Things 1–2 (2018) 67–73 Internet of Things.
- Esmael Hamuda , Brian Mc Ginley, Martin Glavin, Edward Jones, " Improved image processing-based crop detection using Kalman filtering and the Hungarian algorithm", Computers and Electronics in Agriculture 148 (2018).
- M. K. Gayatri, J. Jayasakthi and Dr. G. S. Anandha Mala, "Providing Smart Agriculture Solutions to Farmers for better yielding using IOT," IEEE International Conference on Technological Innovations in ICT for Agriculture and Rural Development, 2015.
- Santanu Phadikar and Jaya Sil "Rice Disease identification using Pattern Recognition Techniques" IEEE Proceedings of 11th International Conference on Computer and Information Technology (ICCIT 2008), Khulna, Bangladesh, pp. 1-4244-2136-7/08, 25-27 December, 2008.

Dheeb Al Bashish, Malik Braik and Sulieman Bani-Ahmad "A Framework for Detection and Classification of Plant Leaf and Stem Diseases" IEEE International Conference on Signal and Image Processing, pp. 978-1-4244-8594-9/10, 2010.

Zulkifli Bin Husin, Abdul Hallis Bin Abdul Aziz, Ali Yeon Bin Md Shakaff and Rohani Binti S Mohamed Farook "Feasibility Study on PlantChili Disease Detection Using Image Processing Techniques" IEEE Third International Conference on Intelligent Systems Modelling and Simulation, pp. 978-0-7695-4668-1/12, 2012.

Sabah Bashir and Navdeep Sharma "Remote Area Plant Disease Detection Using Image Processing" IOSR Journal of Electronics and Communication Engineering (IOSRJECE) ISSN : 2278-2834 Volume 2, Issue 6, PP 31-34, Sep-Oct 2012.

Nonik Noviana kurniawati & Salwani Abdullah, "Texture Analysis for Dignosing Paddy Disease(2009),"

Arnal barbedo & Jayme Garcia, Digital Image Processing Technique for Detecting, Quantifying and Classifying Plant Diseases(2008) .

Shenweizheng & wcyachun, "Grading method of Leaf Spot Disease based on Image Proccessing(2008)," .

Arnal barbedo & Jayme Garcia, Digital Image Processing Technique for Detecting, Quantifying and Classifying Plant Diseases(2008) .